

Named Entities & Responsible NLP

Pre-Assessment 3 · 9 questions · 33 points

Name: _____

Date: _____

Score: / 33

This is a **pre-assessment**. It shows what you already know **before** we start this unit, so it is fine if some questions cover things you have not learned yet. Try every question you can. If you have no idea at all, write **“not yet”** and move on.

Work on your own, without a computer. You have about **40 minutes**. When a question asks what a program prints, write exactly what Python would print, one line of output per answer line. If the code would crash, write the name of the error (for example, `TypeError`). Show your thinking; partial credit is given.

In every spaCy question, assume this code has already run: `import spacy` and `nlp = spacy.load("en_core_web_sm")`

1. Name that entity

5 points

Write the spaCy label for each entity: **PERSON**, **ORG**, **GPE**, **DATE**, or **MONEY**. Each label is used once.

Entity	Label
Serena Williams	
NASA	
Brazil	
July 4th	
\$50	

2. Entity finder

4 points

```
doc = nlp("Maya visited Paris in June with her dad.")
for ent in doc.ents:
    print(ent.text, ent.label_, ent.start_char)
```

(a) Write what the program prints. (spaCy finds three entities here.)

(3 pts)

(b) What does `start_char` tell you?

(1 pt)

3. Entity Gallery

2 points

(a) Write the line that highlights the entities of `doc` in a notebook (`displacy` is already imported).

(1 pt)

(b) Which `displacy` function shows the picture in a web browser instead of a notebook?

(1 pt)

4. When the model is wrong

4 points

Here is what spaCy really found in two sentences.

SPACY OUTPUT

```
Washington crossed the river.  → Washington ORG
i met jordan at apple.         → jordan PERSON
```

(a) In the first sentence, what should the label be, and why did the model get it wrong?

(2 pts)

(b) In the second sentence, which entity did the model **miss**, and why is this sentence harder?

(2 pts)

5. NER Report Card

3 points

These are real spaCy guesses. For each row, write ✓ if the label is right or ✗ if it is wrong. For each ✗, write the correct label (or “none” if it isn't an entity).

Sentence	Entity	Model's label	✓ or ✗	Correct label if ✗
Maya visited Paris in June.	Maya	PERSON		
Washington crossed the river.	Washington	ORG		
Mia ate a taco and Ice Cream in Denver.	taco	GPE		
Ava Stone lives in Chicago.	Chicago	GPE		
Maya visited Paris in June.	June	DATE		

(a) Complete the ✓/✗ column and the corrections (in the table above). (2 pts)

(b) What is the model's score, as a fraction and as a percent? (1 pt)

6. Character Tracker

4 points

```
from collections import Counter
names = ["Dorothy", "Toto", "Dorothy Gale", "Scarecrow", "Dorothy", "Toto"]
aliases = {"Dorothy Gale": "Dorothy"}
clean = [aliases.get(n, n) for n in names]
counts = Counter(clean)
print(counts.most_common(2))
```

(a) What does it print? (2 pts)

(b) Without the `aliases` step, what would `Counter(names) ["Dorothy"]` be? (1 pt)

(c) Write a list comprehension that collects the text of every **PERSON** entity in `doc`. (1 pt)

7. How models learn

3 points

(a) How did spaCy's model learn to recognize names of people and places?

(1 pt)

(b) Suppose most of its training examples were news stories full of American and European names. What might go wrong?

(1 pt)

(c) Name one thing we could do about this problem.

(1 pt)

8. Responsible NLP

3 points

A school wants to use NLP to read every student's essays automatically.

(a) Give one possible **benefit**.

(1 pt)

(b) Give one **privacy or fairness** risk.

(1 pt)

(c) Write one sentence for this tool's **Limitations** card.

(1 pt)

9. Who's Who

5 points

Write a function `people_in(text)` that runs `nlp` on the text and **returns** an alphabetically sorted list of the **different** PERSON names it finds (no repeats).

EXAMPLE: PEOPLE_IN("MAYA AND LEO VISITED PARIS. THEN MAYA MET AVA STONE IN LONDON.") RETURNS

```
['Ava Stone', 'Leo', 'Maya']
```


END OF ASSESSMENT